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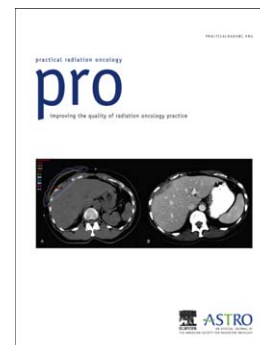
Causal Factors for Error in Radiation Oncology

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Causal Factors for Error in Radiation Oncology

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In this issue of *Practical Radiation Oncology*, the authors of “Evaluation of Near Miss and Adverse Events in Radiation Oncology using a Comprehensive Causal Factor Taxonomy” present their analysis of the contributing factors taxonomy from the AAPM consensus document¹, an adaption of which is presently in use in RO-ILS: Radiation Oncology Incident Learning System[®]. This work is important because it represents an initial exploration of what the causal factors are in radiation oncology from high quality institutional data, as well as international data. It is important to note the differences between the two sources of data used for this analysis: the author’s institutional data, and the international (Safety in Radiation Oncology) SAFRON data. The institutional data is from a center with high levels of event reporting (including near misses, process improvement, unsafe conditions, and incidents) with a healthy feedback loop that supports such high levels of reporting. In contrast, the SAFRON data is an international system, comprised of 42 separate centers from last report,² with varying levels of reporting thresholds, and varying levels of safety culture at each department. Whether all of these 42 centers has an individual incident learning system in which a broader range of reporting occurs is not known. The causal factors for the reported events were determined through a consensus of three skilled raters. Although no inter-rater reliability was presented for these individuals, existing data indicate that the kappa (0.77) for causal factors in this formalism is quite good.³

Given the difference in the datasets, it comes as no surprise that the institutional data has a minority (slightly over 10% for Near Miss Risk Index, NMRI,3-4 group) of actual incidents (reaching the patient), whereas this is the predominant event type (>80%) reported in the SAFRON data. What is of interest, however, is the difference in causal factors between the institutional and SAFRON data. Notably, in the SAFRON database, which is made up nearly entirely of incidents (reaching the patient), inadequate policy is double what is in the single institution data. Likewise, the presence of human behavior causal factors (slip, poor judgment) is also at least double in the SAFRON data. It is not clear that this represents an intrinsic difference between events not reaching the patients and actual incidents. However, it is tempting to speculate that this may be a benefit of high-level incident learning systems. One can imagine that in a high volume reporting system, learning from the “free lessons”^{4,5} of near misses allows existing policies to be bolstered, resulting in a decrease in “inadequate policy” as a causal factor. Likewise, one will remember the words: “We cannot change the human condition, but we can change the conditions under which humans work.”⁶ What are we to make of the decreased rates of “slips” in the institutional data versus the SAFRON data? Is this merely reflective of the nature of incidents versus events not reaching the patient? Or could it be that the “conditions under which humans work” is more favorable in institutions in which engage heavily in incident learning and continuous quality improvement, resulting in less permeability to error from human behavior issues? We can be confident that the therapists, physicists, and physicians are equally devoted to quality patient care in both systems. We cannot know from this data what exactly is responsible for this difference, as it is not possible to understand exactly the level to which SAFRON participating institutions engage in incident learning outside of the SAFRON system. Nonetheless, this data is thought provoking.

Another notable lesson from this data is the distribution of poor human factors engineering as a causal factor. In the institutional data, high NMRI events list this as a causal factor more than twice as often as in low NMRI events. This is one of the most striking things from this analysis. The implication that events with significant risk of harm are associated with human factors engineering issues is, to me, good news. This is something concrete that we can fix, with the help of our vendor colleagues. The RO-ILS team has obtained financial support from existing radiation oncology vendors, which signifies an initial commitment to supporting the procurement of this data. Members of the Radiation Oncology Healthcare Advisory Council, the team that reviews data entered into RO-ILS, presented data and opportunities for improvements to high-level vendor supporters of RO-ILS (personal communication, Cindy Tomlinson, Ksenija Kapetanovic). Although the SAFRON data had a much lower rate of identified causal factors citing human factors engineering, I would argue this is an artifact of the limited information in the SAFRON reports. Without intrinsic knowledge of which platforms were being used, and how the data display impacts interpretation and interface with the system, it is unlikely that the raters had adequate information to identify human factors engineering issues within the SAFRON data.

This data also gives support to the nature of error in radiation therapy. A median of seven causal factors was selected for each event, and no event had one causal factor. This is important confirmation of what we all know about error: the swiss cheese model is real.⁶ It is not a bad apple, it is not one person, one thing, or one “root cause.” Error happens because of system vulnerabilities that allow permeability of the system to error. This data gives further support to the systems view of error, and supports the practice of thorough investigation techniques.⁷

Additionally, the raters identify four additional causal factors not fully addressed in the current taxonomy: machine down time, hardware/equipment failure, failure to follow through with a quality improvement intervention, and misleading workflow documentation. I would argue that there is another potential for expanding the current taxonomy: the arena of cognitive bias. The joint commission (TJC) has implicated cognitive bias as a significant problem underlying errors in medicine.⁸ There are over 100 identified cognitive biases in the literature.^{9,10} Despite this, the current RO-ILS taxonomy has a single cognitive bias listed: expectation bias. I would argue that this is a rather anemic list, and we should consider expanding this in future versions of the taxonomy. TJC has also identified a series of factors which make cognitive bias more likely, which could be an opportunity to modulate this aspect of error. These include person factors (cognitive loading, fatigue, affective bias), patient factors (complexity, poor history, comorbidities), and system factors (task complexity, handoffs, insufficient time, inadequate processes, distractions, poorly functioning information technology, poor teamwork, etcetera). Additionally, cognitive forcing strategies exist to help counteract the influence of these tendencies.^{11,12} Hopefully future versions of this taxonomy may be able to incorporate bias more fully to that its contribution to error in radiation oncology may be more fully realized.

I am pleased to see this contribution to the field of error management, and I applaud the authors on their thoughtful consideration of this subject.

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